BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 **Data Analysis Report**

Project ID: R26-DS-004

Project Title: An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka

1.Introduction

1.1 Background

Sri Lankan taxpayers often find it difficult to understand and comply with income tax regulations because the tax system includes complex rules, multiple income categories, deductions, relief schemes, and legal requirements defined under the Inland Revenue Act No. 24 of 2017 and its amendments. Due to this complexity, many individuals depend on auditors or tax consultants to calculate taxes, identify possible tax-saving opportunities, and ensure compliance with tax laws. However, this manual process can be expensive, time-consuming, and challenging for people who have limited financial or tax knowledge.

Even though online tax filing systems and digital tax platforms are available today, most of these systems mainly support tax calculation and return submission. They do not provide intelligent or personalized tax advice based on an individual's financial situation. In addition, existing systems rarely explain why a particular tax recommendation is given, making it difficult for users to fully understand their tax decisions. Many AI-based financial systems also work like "black boxes," where users cannot clearly see how recommendations are generated, reducing transparency and trust.

With recent developments in Artificial Intelligence (AI), Machine Learning (ML), Explainable AI (XAI), and Natural Language Processing (NLP), there is an opportunity to improve digital tax advisory services. These technologies can help analyze financial information, apply tax regulations intelligently, recommend suitable tax-saving strategies, and provide understandable explanations for users.

Therefore, this research proposes an Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka. The proposed system combines financial transaction analysis, tax strategy optimization, intelligent tax advisory models, and personalized recommendation mechanisms to provide explainable, personalized, and legally compliant tax guidance for Sri Lankan taxpayers.

1.2 Research Problem

In Sri Lanka, many taxpayers struggle to understand and comply with income tax regulations because the taxation system is complex and involves multiple income categories, deductions, exemptions, and relief schemes under the Inland Revenue Act No. 24 of 2017 and its amendments. Calculating taxes and identifying the most suitable tax-saving opportunities often require professional knowledge and careful interpretation of tax laws. Due to this complexity, individuals frequently rely on auditors, accountants, or tax consultants for tax calculation, planning, and compliance support. However, this manual approach can be expensive, time-consuming, and difficult to access for many taxpayers, especially individuals with limited financial or tax knowledge.

Although several digital tax systems and online filing platforms are available today, most existing systems mainly focus on tax calculation and return submission rather than helping users make informed tax planning decisions. These systems usually do not provide personalized recommendations based on a taxpayer's financial profile, future financial goals, or behavioral patterns. In addition, many existing solutions lack transparency and explainability, making it difficult for users to understand why a particular recommendation or tax calculation was generated. This reduces user trust and limits the effectiveness of automated tax advisory systems.

Furthermore, there is limited research and very few intelligent tax advisory systems specifically designed for the Sri Lankan taxation context. Existing AI-based financial systems are often developed for other countries or focus mainly on auditing, fraud detection, or basic financial forecasting rather than personalized tax optimization and explainable decision support.

Therefore, the main research problem addressed in this study is the lack of an intelligent, explainable, and personalized tax advisory system that can analyze financial information, identify legal tax-saving strategies, provide understandable recommendations, and support taxpayers in making informed tax planning decisions while ensuring compliance with Sri Lankan tax regulations.

1.3 Objectives:

IT Number	Objective	Objective number
IT22118936	Develop a Financial Transaction Semantic Reasoning and Taxable Income Inference module to analyze financial transaction data and identify taxable income categories using explainable AI techniques.	1
IT22064486	Develop a Tax Strategy Optimization and Explainable Decision Engine to generate personalized and legally compliant tax-saving strategies for Sri Lankan taxpayers.	2
IT22896186	Develop an Intelligent Tax Advisory Language Model to provide explainable, citation-based, and legally grounded tax advisory support using Sri Lankan tax regulations and IRD documents.	3
IT22238580	Develop a Personalized Recommendation and Predictive Impact Modeling Engine to rank tax-saving strategies, predict future financial impact, and support personalized tax planning decisions.	4

2.Data Exploration

2.1 Data Collection

Data for this research was collected with the support of Ernst & Young (EY) Sri Lanka. After we presented our research idea and explained the overall system we planned to develop, the EY team reviewed the project and agreed to support us throughout the research process. They provided valuable domain knowledge related to Sri Lankan taxation, financial auditing, and real-world tax compliance processes, which helped us better understand practical industry challenges and requirements.

To support the research, EY shared relevant financial transaction samples and tax-related documents in both soft-copy and hard-copy formats. Since financial information is sensitive, all user-specific and personal details were removed before sharing the data with us. The data was provided in an anonymized and de-identified manner to maintain privacy and confidentiality while still allowing us to use realistic financial information for analysis and validation purposes.

In addition to the data provided by EY, we also collected publicly available Inland Revenue Department (IRD) regulations, tax circulars, and legal documents to support the tax advisory and rule-based reasoning components of the system. Furthermore, synthetic taxpayer datasets and benchmark datasets were generated to simulate realistic Sri Lankan taxpayer scenarios for machine learning model training, testing, and evaluation.

2.2 Dataset Description

Data Source	Description	Resource	Size	Key Attributes
Financial Transaction Dataset	Anonymized bank transaction records used for taxable income inference and transaction semantic analysis	EY Sri Lanka + Synthetic Data	20,000+ records	Transaction description, amount, date, CR/DR type, bank details
Synthetic Taxpayer Financial Dataset	Artificially generated taxpayer financial profiles for recommendation and optimization modeling	Generated Dataset	25,000+ records	Income, expenses, deductions, EPF/ETF, liabilities, investments
Tax Calculation Dataset	Dataset containing calculated tax values based on different income levels, deductions, and tax relief schemes under Sri Lankan tax regulations	Rule-Based Generated Dataset	18,000+ records	Gross income, taxable income, deductions, tax payable, reduced tax amount
Tax Reduction & Strategy Dataset	Dataset containing different legal tax-saving strategies and their financial impact	Synthetic + Rule-Based Dataset	15,000+ records	Strategy type, estimated savings, reduced tax, adoption probability, risk level
IRD Legal and Tax Corpus	Sri Lankan Inland Revenue regulations, tax circulars, and legal documents for advisory and rule-based reasoning	IRD Sri Lanka Publications	1000+ documents/chunks	Tax rules, legal references, reliefs, tax categories

Tax Query & Intent Dataset	Tax-related user queries and benchmark intent dataset for advisory language model training	Generated Dataset	5,000+ queries	Tax queries, intent labels, advisory categories
Recommendation & Ranking Dataset	Dataset used for ranking personalized tax-saving recommendations and predictive modeling	Synthetic Dataset	12,000+ records	User profile, ranking score, financial behavior, recommendation priority
Explainability & SHAP Dataset	Dataset generated for explainable AI analysis and feature contribution evaluation	ML Generated Dataset	10,000+ records	Feature importance, SHAP values, prediction contribution, confidence score

2.3 Suitability Analysis

2.3.1 Relevance to Individual Research Objectives:

Data Source	Objective 1	Objective 2	Objective 3	Objective 4
Financial Transaction Dataset	Used for semantic transaction analysis and taxable income inference	Supports tax optimization using categorized income data	Provides transaction context for tax advisory responses	Supports personalized financial behavior analysis
Synthetic Taxpayer Financial Dataset	Supports taxable income classification and feature extraction	Used for strategy optimization and recommendation modeling	Provides user financial context for advisory generation	Used for personalized recommendation and prediction
Tax Calculation Dataset	Helps identify taxable and non-taxable income categories	Supports tax calculation and optimization evaluation	Provides tax calculation references for advisory responses	Used to predict financial impact and reduced tax outcomes
Tax Reduction & Strategy Dataset	Supports mapping of tax-related financial patterns	Used for strategy ranking and optimization	Provides recommendation context for advisory explanations	Supports recommendation ranking and future impact prediction
IRD Legal and Tax Corpus	Supports tax-rule mapping and reasoning	Used for legal validation of tax-saving strategies	Main knowledge source for the advisory language model	Supports explainable recommendation generation

Tax Query & Intent Dataset	Supports transaction-related query understanding	Helps identify optimization-related user intents	Used for intent classification and retrieval tasks	Supports personalized query recommendations
Recommendation & Ranking Dataset	Supports financial behavior understanding	Used for recommendation ranking and optimization	Provides ranked outputs for advisory explanations	Main dataset for personalized recommendation modeling
Explainability & SHAP Dataset	Supports explainable income inference	Explains optimization and ranking decisions	Supports explainable advisory outputs	Used for transparent recommendation and prediction analysis

3. Methodology

3.1 Data Preprocessing

Objective 1- Financial Transaction Semantic Reasoning and Taxable Income Inference

Data Source	Data Cleaning							
Financial Transaction Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Synthetic Taxpayer Dataset	✓	✓	✓	✓	✓	✓		✓
Tax Calculation Dataset	✓	✓		✓	✓	✓		✓
IRD Legal & Tax Corpus	✓	✓			✓			✓
Tax Query & Intent Dataset	✓	✓	✓	✓	✓			✓

Objective 2 : Tax Strategy Optimization and Explainable Decision Engine

Data Source	Data Cleaning	Data Normalization	Data Encoding	Missing Data Handling	Feature Engineering	Outlier Handling	Data Scaling	Data Mapping
Synthetic Taxpayer Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Tax Calculation Dataset	✓	✓		✓	✓	✓	✓	✓
Tax Reduction & Strategy Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Recommendation & Ranking Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Explainability & SHAP Dataset	✓	✓		✓	✓		✓	✓
IRD Legal & Tax Corpus	✓	✓			✓			✓

Objective 3: Intelligent Tax Advisory Language Model

Data Source	Data Cleaning	Data Normalization	Data Encoding	Missing Data Handling	Feature Engineering	Text Processing	Data Mapping	Data Integration
IRD Legal & Tax Corpus	✓	✓		✓	✓	✓	✓	✓
Tax Query & Intent Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Financial Transaction Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Recommendation & Ranking Dataset	✓	✓	✓	✓	✓		✓	✓
Synthetic Taxpayer Dataset	✓	✓	✓	✓	✓		✓	✓
Explainability & SHAP Dataset	✓	✓		✓	✓		✓	✓

Objective 4: Personalized Recommendation and Predictive Impact Modeling

Data Source	Data Cleaning	Data Normalization	Data Encoding	Missing Data Handling	Feature Engineering	Outlier Handling	Data Scaling	Data Mapping
Recommendation & Ranking Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Synthetic Taxpayer Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Tax Reduction & Strategy Dataset	✓	✓	✓	✓	✓	✓	✓	✓
Tax Calculation Dataset	✓	✓		✓	✓	✓	✓	✓
Explainability & SHAP Dataset	✓	✓		✓	✓		✓	✓
Financial Transaction Dataset	✓	✓	✓	✓	✓	✓	✓	✓

3.2 Scalability

The datasets used in this research were designed to be sufficiently large, diverse, and scalable to support machine learning, explainable AI, recommendation systems, and language model-based analysis. Since obtaining large-scale real-world taxpayer data is difficult due to privacy and confidentiality concerns, synthetic datasets were generated to simulate realistic Sri Lankan taxpayer scenarios while maintaining data diversity and balance across different income categories, deductions, tax-saving strategies, and financial behaviors.

The synthetic taxpayer financial dataset contains more than 25,000 records and includes various demographic, financial, and tax-related attributes such as income levels, deductions, liabilities, investments, EPF/ETF balances, and risk preferences. This dataset is large enough to train and evaluate recommendation models, ranking algorithms, and optimization techniques effectively.

The financial transaction dataset contains over 20,000 anonymized transaction records with multilingual transaction descriptions, transaction amounts, dates, and transaction types. This dataset supports scalable semantic transaction analysis and taxable income inference tasks. In addition, IRD legal documents, tax circulars, and regulatory datasets were structured into chunk-level corpora to support scalable retrieval, advisory, and knowledge-based reasoning processes within the Intelligent Tax Advisory Language Model.

The overall system architecture was designed using modular FastAPI services and scalable machine learning pipelines, allowing datasets and models to be expanded in the future with additional taxpayer records, financial transactions, or updated tax regulations. Therefore, the current dataset sizes are considered sufficient for research experimentation, model training, validation, and future scalability improvements.

3.3 Feature extraction

Feature extraction was carried out to identify the most important attributes and patterns required for each research objective and to improve the performance of the machine learning and AI models used in the system. Since the research includes multiple components such as transaction analysis, tax optimization, advisory generation, and recommendation modeling, different types of features were extracted from each dataset based on the specific requirements of the component.

For the Financial Transaction Semantic Reasoning component, important transaction-related details such as transaction descriptions, transaction amounts, transaction type (CR/DR), bank details, and transaction dates were extracted from the financial transaction datasets. Transaction descriptions were further processed using Natural Language Processing (NLP) techniques to identify meaningful patterns and categorize income-related transactions more accurately.

For the Tax Strategy Optimization and Personalized Recommendation components, several financial and behavioral attributes were extracted from taxpayer financial profiles. These included income levels, expenses, deductions, EPF/ETF balances, liabilities, investments, risk tolerance, and financial behavior patterns. Additional features such as estimated tax liability, disposable income, effective tax rate, and possible tax reduction values were calculated using Sri Lankan tax rules and rule-based computations.

For the Intelligent Tax Advisory Language Model, important textual and legal features were extracted from IRD regulations, tax circulars, and tax-related user queries. Features such as intent labels, legal references, keywords, tax categories, and section mappings were identified to support intelligent query understanding and citation-based advisory generation.

In addition, recommendation scores, adoption probabilities, estimated savings, and ranking-related attributes were extracted for the recommendation and predictive modeling components. Explainability-related features such as feature importance and SHAP values were also generated to help users understand how recommendations and predictions were produced by the system.

4. Modelling and Results

This section presents the key findings and outcomes obtained through data analysis, machine learning model development, and evaluation processes carried out for the proposed AI-driven tax advisory system. Different modelling techniques, including machine learning, rule-based reasoning, explainable AI, recommendation systems, and retrieval-based language models, were applied to address the research objectives and improve personalized tax compliance and advisory support for Sri Lankan taxpayers.

For the Financial Transaction Semantic Reasoning component, machine learning and NLP-based models were used to classify financial transactions and identify taxable income categories from raw transaction descriptions. The analysis showed that transformer-based multilingual models performed effectively in understanding transaction semantics and improving classification accuracy, especially for mixed-language transaction descriptions commonly found in Sri Lankan banking records.

For the Tax Strategy Optimization and Personalized Recommendation components, rule-based tax engines and machine learning ranking models were used to generate and evaluate legal tax-saving strategies. The models successfully identified optimized tax-saving opportunities based on user financial profiles while ensuring compliance with Sri Lankan tax regulations. Recommendation and ranking models also demonstrated the ability to predict suitable tax-saving strategies and estimate potential financial impact and reduced tax outcomes for taxpayers.

For the Intelligent Tax Advisory Language Model, retrieval-based NLP techniques and knowledge-grounded advisory mechanisms were used to generate explainable tax guidance using IRD regulations and legal references. The results showed improved transparency and reliability compared to general AI-generated responses, as the system provided citation-based and legally grounded advisory outputs.

Explainable AI techniques such as SHAP were integrated into the recommendation and optimization components to improve user understanding and transparency of model predictions. Overall, the analysis and modelling results demonstrated that the proposed system can provide intelligent, personalized, explainable, and legally compliant tax advisory support while reducing the dependency on manual tax consultation processes.

4.1 Key Insights

The data analysis process provided several useful insights about taxpayer financial behavior, tax-saving patterns, financial transactions, and personalized tax recommendations. Different datasets were analyzed using statistical analysis, machine learning techniques, and visualization methods such as charts, graphs, correlation heatmaps, and distribution plots to better understand the relationships between financial factors and tax-related outcomes.

From the synthetic taxpayer financial dataset, it was observed that income level, deductions, liabilities, and investments have a strong impact on tax liability and tax-saving opportunities. Higher-income individuals generally showed greater use of deductions and investment-related tax benefits, while middle-income taxpayers were more affected by expenses, debt obligations, and disposable income levels. The analysis also showed a clear relationship between income, deductions, and the final reduced tax amount.

The financial transaction dataset revealed that many transaction descriptions follow recurring patterns which can be used to identify taxable and non-taxable income categories. By applying NLP-based preprocessing and classification techniques, the system was able to better understand transaction meanings and improve taxable income identification. Transaction frequency analysis and category distribution graphs also helped identify common taxpayer income sources and spending behaviors.

The recommendation and optimization datasets showed that personalized tax-saving recommendations are more effective than generic recommendations. Financial behavior, risk tolerance, investment patterns, and deduction usage significantly influenced the ranking of tax-saving strategies. Visualizations such as savings comparison charts and ranking graphs helped identify which strategies provided the highest potential tax reduction for different taxpayer profiles. For the Intelligent Tax Advisory Language Model, the analysis showed that citation-based responses using IRD regulations improved user trust and transparency compared to general AI-generated answers. Query analysis also highlighted that taxpayers frequently seek guidance related to deductions, tax filing procedures, eligibility criteria, and tax calculation methods.

In addition, explainability analysis using SHAP values helped identify the features that most influenced prediction and recommendation outputs. Features such as taxable income, deduction amount, investment behavior, and debt-to-income ratio had a strong impact on recommendation rankings and tax optimization results. Overall, the analysis demonstrated that combining AI techniques, rule-based tax reasoning, and explainable recommendations can significantly improve personalized tax advisory and decision support for Sri Lankan taxpayers.

4.2 Challenges Faced During Data Analysis:

Several challenges were encountered during the data collection, preprocessing, and analysis stages of this research. One of the main challenges was the limited availability of real-world taxpayer and financial transaction data due to privacy, confidentiality, and legal restrictions. Since financial information is highly sensitive, most of the datasets had to be generated synthetically or provided in an anonymized format, which required additional effort to ensure the generated data realistically represented Sri Lankan taxpayer scenarios.

Another challenge was handling inconsistent and unstructured financial transaction descriptions. Many transaction records contained abbreviations, mixed-language text, incomplete descriptions, and different banking formats, making transaction classification and semantic understanding more difficult. Additional preprocessing and NLP-based normalization techniques were required to improve the quality and consistency of the data.

Missing values and incomplete financial records were also identified in several datasets. Some records lacked important attributes such as deduction details, investment information, or transaction categories. Different missing data handling techniques such as imputation, default value assignment, and record filtering were used to address these issues while maintaining dataset quality.

Outliers and extreme financial values presented another challenge during analysis. Some generated financial profiles included unusually high income levels, deductions, or investment values that could negatively affect model training and recommendation accuracy. Outlier detection and normalization techniques were applied to reduce the impact of such irregularities.

The research also faced challenges related to multilingual financial and legal text processing. Transaction descriptions and user tax queries sometimes included a mixture of English and Sinhala terms, making NLP-based feature extraction and semantic analysis more complex. Additional preprocessing and multilingual language handling techniques were required to improve classification and retrieval performance.

In addition, computational limitations were experienced during machine learning model training, especially for transformer-based NLP models and recommendation ranking models. Large datasets and explainability calculations such as SHAP analysis required significant processing power, memory usage, and training time. Efficient preprocessing pipelines and optimized model configurations were used to manage these computational challenges.

Despite these challenges, suitable preprocessing methods, scalable architectures, synthetic data generation techniques, and explainable AI approaches helped ensure that the datasets and models remained reliable and effective for achieving the research objectives.

5. References

[1] Inland Revenue Department of Sri Lanka, *Inland Revenue Act No. 24 of 2017*. Colombo, Sri Lanka: IRD Sri Lanka, 2017.

[2] Inland Revenue Department of Sri Lanka, "Tax Circulars and Public Guidelines," IRD Sri Lanka. [Online]. Available: https://www.ird.gov.lk. [Accessed: 13-May-2026].

[3] Ernst & Young Sri Lanka, "Industry Guidance and Anonymized Financial Data Support for Research Validation," Colombo, Sri Lanka, 2026.

[4] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794.

[5] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, Long Beach, CA, USA, 2017, pp. 4765–4774.

[6] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, Minneapolis, MN, USA, 2019, pp. 4171–4186.

[7] Y. Liu *et al.*, "RoBERTa: A Robustly Optimized BERT Pretraining Approach," arXiv preprint arXiv:1907.11692, Jul. 2019.

[8] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[9] Microsoft, "LightGBM Documentation." [Online]. Available: https://lightgbm.readthedocs.io. [Accessed: 13-May-2026].

[10] FastAPI, "FastAPI Framework Documentation." [Online]. Available: https://fastapi.tiangolo.com. [Accessed: 13-May-2026].

[11] Hugging Face, "Transformer Models and NLP Libraries." [Online]. Available: https://huggingface.co. [Accessed: 13-May-2026].

[12] Neo4j, "Neo4j Graph Database Platform." [Online]. Available: https://neo4j.com. [Accessed: 13-May-2026].

[13] LangChain, "LLM Application Framework." [Online]. Available: https://www.langchain.com. [Accessed: 13-May-2026].

[14] Department of Census and Statistics Sri Lanka, *Household Income and Expenditure Survey Report 2024*. Colombo, Sri Lanka, 2024.

[15] C. Molnar, *Interpretable Machine Learning*, 2nd ed. 2022. [Online]. Available: https://christophm.github.io/interpretable-ml-book/. [Accessed: 13-May-2026].

[16] User Financial Profiles synthatic dataset

[17] Transaction Sementic Synthetic Dataset

[18] Tax calculation synthetic dataset